

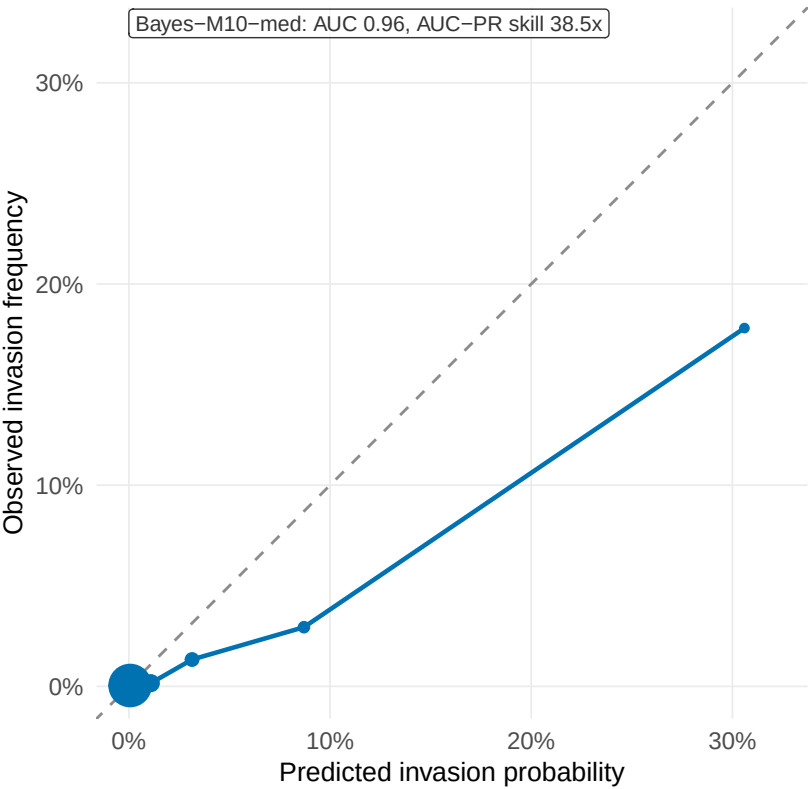
How good is the invasion model? (h = 1w)

(A) calibration + AUC and (B) real-time prioritisation vs random – after Kraemer & Cauchemez 2017, Lancet Inf Dis, Fig 3

(A) Calibration: predicted vs observed invasion (h = 1w)

On the diagonal = calibrated probabilities; below = over-confident.

zone-weeks ● 1000 ● 2000 ● 3000 ● 4000 —●— Bayes-M10-med

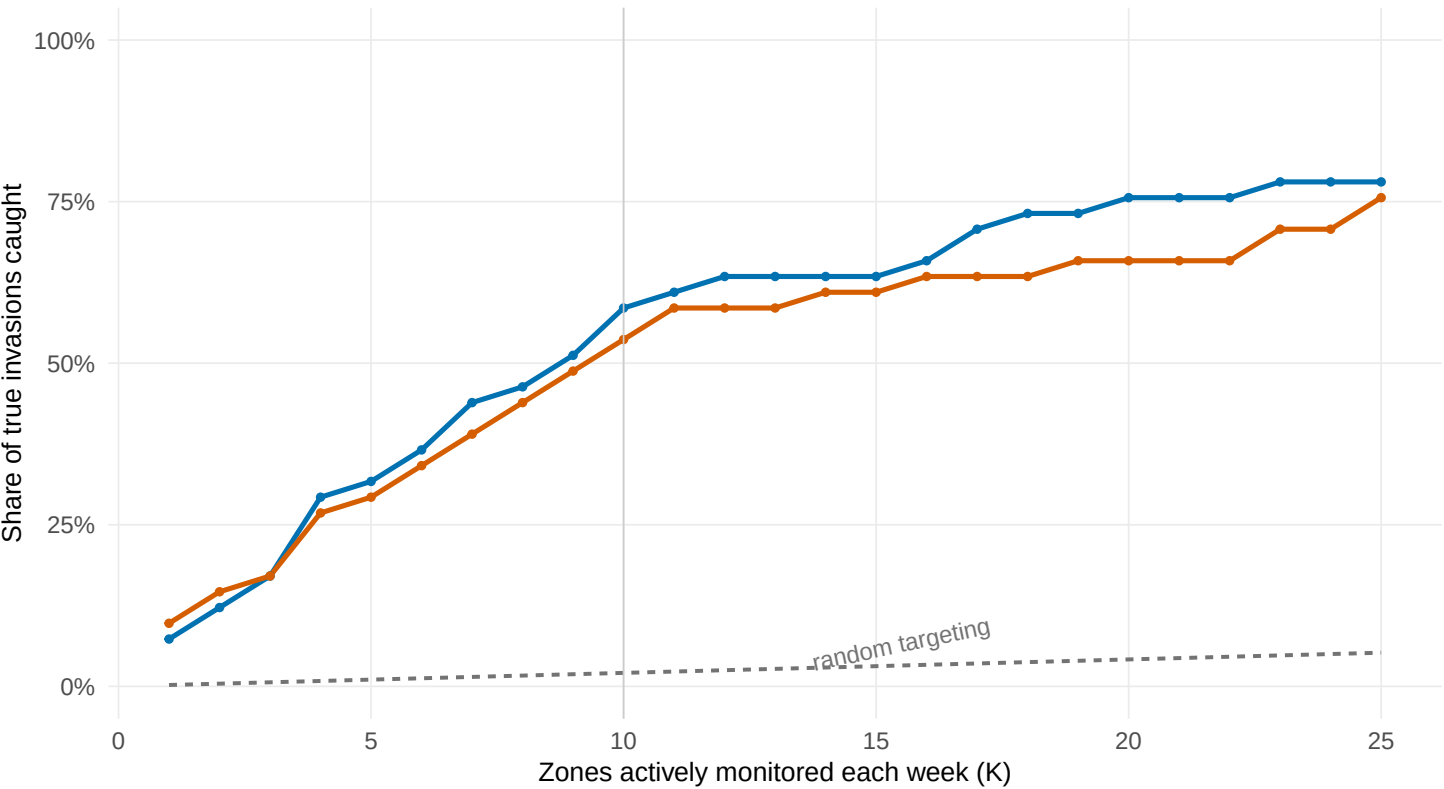


The 2.3% of at-risk zone-weeks we score >15%/week had a 18% observed invasion rate.

(B) Prioritisation: invasions caught vs random targeting

Share of true next-week invasions caught at a fixed weekly alert budget of K zones, pooled over LFO folds

—●— Bayes-M10-med —●— Naive-epicentre-inflow



At a budget of K=10 zones/week, Bayes-M10-med catches 59% of the next invasions vs 2% for random targeting (28.1x better).